

A Universal Attenuation Model of Terahertz Wave in Space-Air-Ground Channel Medium

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ABSTRACT Providing continuous bandwidth over several tens of GHz, the Terahertz (THz) band (0.1-10 THz) supports space-air-ground integrated network (SAGIN) in 6G and beyond wireless networks. However, it is still mystery how THz waves interact with the channel medium in SAGIN. In this paper, the absorption and scattering effect in THz space-air-ground channels are modeled as a collision process between THz photons and medium particles, based on which a universal space-air-ground attenuation model is proposed for THz waves. The proposed model incorporates the attenuation effects induced by particles including condensed particles, molecules, and free electrons. The propagation loss of THz waves in the atmosphere and the outer space are numerically assessed. Numerical results indicate that the attenuation effects except free space loss are all negligible at the altitude higher than 50 km while they need to be considered in the atmosphere lower than 50 km.

INDEX TERMS Terahertz communications, space-air-ground integrated network (SAGIN), channel modeling.

I. INTRODUCTION

SPACE-AIR-GROUND integrated network (SAGIN), as an emerging communication scenario to provide global coverage for 6G wireless networks, puts a stringent requirement for the high data rates of wireless communications [1], [2], [3], [4]. Providing several tens of Gigahertz (GHz) continuous bandwidth, the Terahertz (THz) band (0.1-10 THz) has been envisioned as one of the key solutions to support the Terabit-per-second (Tbps) data traffic in future 6G and beyond wireless networks [5], [6], [7]. Therefore, THz communications have the potential to realize ubiquitous and global link establishment for future applications and technologies, such as ubiquitous artificial intelligence, industrial Internet of Things, and digital twins [8], [9], [10].

As the first step to establish the THz SAGIN, careful analysis on the THz SAGIN channels, or more specifically, the propagation loss caused by various attenuation effects, are necessary. Compared with lower frequency bands, new

observations on THz channels are summarized as follows. First, according to Friis' law, the free-space path loss (FSPL) of THz wave is higher due to the increased frequency. Second, the absorption effect led by water vapor is significant, which is negligibly weak at lower frequency bands. Third, particles like cloud droplets induce severer attenuation and weaker diffraction for THz waves. Finally, THz waves suffer from weaker attenuation in plasma due to the higher frequency, creating new opportunity to transmit in the ionosphere and solar wind [11]. Thus, it is imperative to completely and precisely evaluate the propagation loss of THz space-air-ground links such that the requirements on link establishment and the feasibility of THz SAGIN can be clarified.

In the literature, existing attenuation models for the THz wave propagation only characterize a certain portions of attenuation effects. For example, as listed in TABLE 1, the attenuation effects in the THz band include Mie absorption and scattering, Rayleigh absorption and scattering,

TABLE 1. Attenuation effects on THz waves in space-air-ground channel medium (d : diameter, λ : wavelength).

Particles	Condensed particles $d \approx \lambda$		Condensed particles $d \ll \lambda$		Molecules		Free electrons		Free-space path loss
	Mie absorption	Mie scattering	Rayleigh absorption	Rayleigh scattering	Molecular absorption	Rayleigh scattering	Coulomb absorption	Thomson scattering	
Troposphere	✓	✓	✓	✓	✓	✓	×	×	✓
Stratosphere	×	×	×	×	✓	✓	×	×	✓
Ionosphere	×	×	×	×	×	×	✓	✓	✓
Outer space	×	×	×	×	×	×	✓	✓	✓

molecular absorption and scattering, Coulomb absorption, and Thomson scattering [12], [13]. However, there exists no universal attenuation models considering all these attenuation effects to characterize THz wave propagation from space, air to ground or vice versa [14]. For instance, in the recommendations of international telecommunication union (ITU), there are three distinct attenuation models, namely ITU-R P.676-13 for molecular absorption loss in the atmosphere [15], ITU-R P.840-8 for the attenuation caused by cloud and fog [16], and ITU-R P.838-3 for the attenuation induced by rain [17]. However, they are all empirical models and remain some problems including:

- ITU-R P.676-13 does not show a perfect accordance with a THz channel model incorporating molecular absorption [18], which utilizes the data from the high resolution transmission molecular absorption database (HITRAN) [15].
- ITU-R P.840-8 does not consider the altitude dependency of cloud or fog attenuation, which limits its application in THz SAGIN with varying altitudes.
- ITU-R P.838-3 does not incorporate the impact of different rain spectra, i.e., the distributions of raindrop diameters. However, as pointed out by [19], different rain spectra can lead to different rain attenuation of THz wave propagation.

For system-level analysis, the absorption loss of THz waves in plasma is modeled in [11], which is utilized to assess the capacity of inter-satellite links. A conventional THz aerial channel model considers both the molecular absorption effect and the loss caused by cloud [20]. In summary, a universal model that incorporates all kinds of effects in SAGIN is still missing.

To address the aforementioned challenges, this paper proposes a universal THz space-air-ground attenuation model based on quantum theory, electrodynamics, and thermodynamics. First, according to quantum theory, the absorption and scattering effects are revealed as the consequences of collision between THz photons and medium particles. As a result, the attenuation effect depends on the absorption and scattering cross sections of each kind of mediums particle and the particle distribution along the propagation path. Second, based on electrodynamics, the cross sections for different diameters of medium particles and wavelengths are classified and characterized. The particle distribution is analyzed for different environmental factors such as altitude

and extreme weather [21]. Finally, the comparison between the proposed model and the ITU models are presented.

The contributions of this work are summarized as follows:

- We propose a universal attenuation model to characterize the collision between photons and particles for THz ground-air-space communications, which incorporates the common attenuation effects on THz waves in the atmosphere, namely, Mie absorption and scattering, Rayleigh absorption and scattering, molecular absorption and scattering, Coulomb absorption, and Thomson scattering.
- We model the spatial distribution of particles in the atmosphere and evaluate the propagation loss in various effects in different layers of the atmosphere and the outer space and different weather conditions.
- We compare and contrast the proposed model and the ITU models. For rain attenuation, the proposed model can incorporate the environment-dependent rain spectra, while the empirical model in ITU-R P.838-3 does not consider the different rain spectra in various scenarios. For molecular absorption effect led by water vapor, the proposed model fits well with ITU-R P.676-13, which verifies the reliability of the proposed model.

The rest of this paper is presented as follows. The universal attenuation model of THz wave in space-air-ground channel medium based on the collision process between THz photons and EM waves is elaborated in Section II. Then, based on the proposed attenuation model, the numerical results on the total propagation loss of different layers of the atmosphere and the comparison with ITU models are assessed in Section III. Finally, Section IV concludes the paper.

Notation: Operator d represents the differential operation. d stands for the diameter of particle. $\text{Re}(\cdot)$ returns the real part of a complex number. $(\cdot)'$ refers to the derivative. $\exp(\cdot)$ defines the exponential function.

II. SPACE-AIR-GROUND ATTENUATION MODEL

During the propagation of THz waves in space-air-ground channels, THz waves suffer from different attenuation effects in addition to FSPL as summarized in Fig. 1. Various components in the channel medium from the troposphere to the outer space, such as raindrops, fog drops, atmospheric molecules, free electrons, etc., induce multifarious attenuation effects on THz waves. In this section, a universal attenuation model for THz wave propagation is analyzed to comprehensively incorporate various attenuation effects, including Mie absorption and scattering, Rayleigh absorption

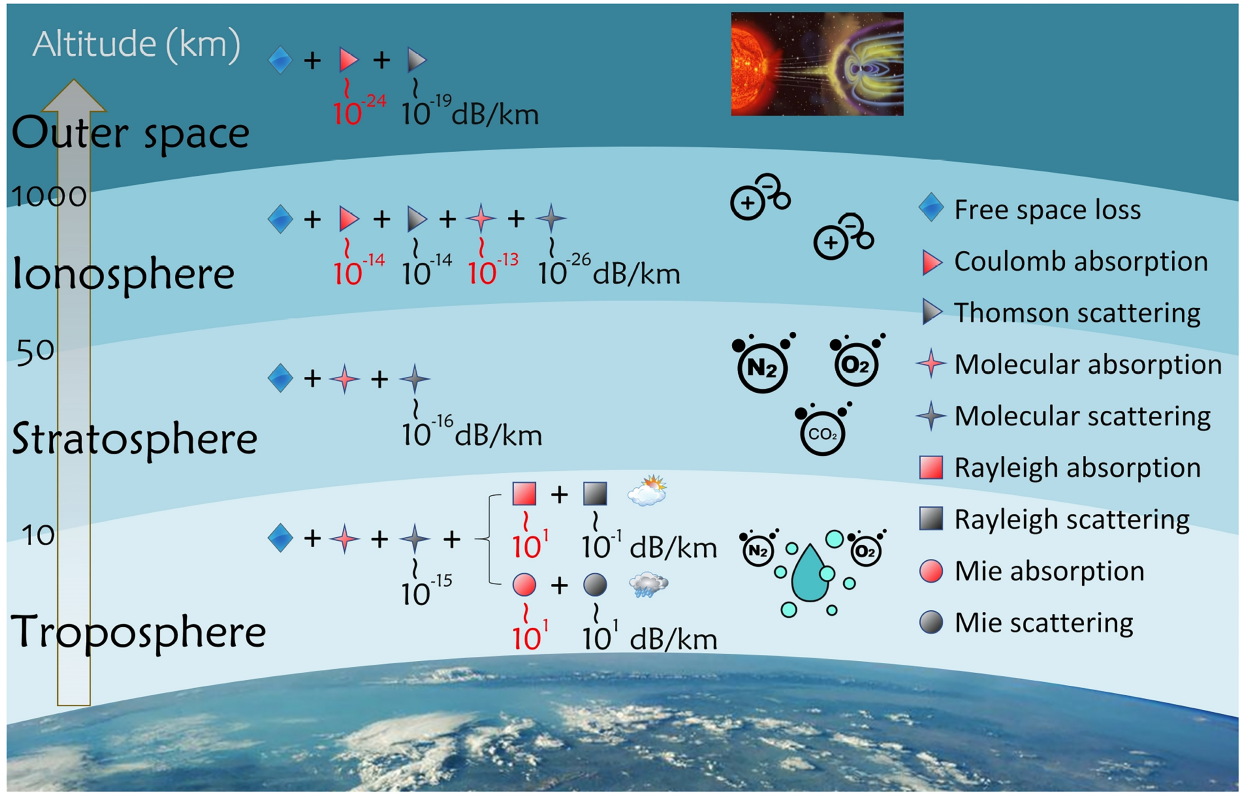


FIGURE 1. THz wave propagation in the atmosphere and the outer space.

and scattering, molecular absorption and scattering, Coulomb absorption, and Thomson scattering. Based on the proposed model, the total propagation loss in space-air-ground channel can be derived for various atmospheric layers and extreme weathers.

A. COLLISION-BASED UNIVERSAL ATTENUATION MODEL

According to wave-particle duality, the attenuation effects of THz wave result from the collisions between THz photons and medium particles along the propagation path of THz space-air-ground channels. To model this collision process, we consider a cylindrical volume along the propagation path with cross section A and length ds . We assume that the particles in this small volume are identical, i.e., the chemical composition and the particle size are uniform. The cross section of a particle in this area, denoted by σ , is defined as the area through which the THz photon travels can collide with the particle. Based on this geometric relationship, the probability that a THz photon collides with a particle inside the cylindrical volume can be expressed by

$$\text{Pr} = \frac{\sigma N}{A}, \quad (1)$$

where N denotes the number of particles in this area. Let n denote the number density of the particles, i.e., the number of particles per unit volume, as

$$n = \frac{N}{A ds}. \quad (2)$$

When a collision happens, the incident photon can be absorbed or scattered by the particle. Therefore, we further divide the cross section into two cross sections σ_A and σ_S , as absorption cross section and scattering cross section of the particle, i.e., $\sigma = \sigma_A + \sigma_S$. By substituting (2) into (1), the collision probability can be expressed as

$$\text{Pr} = n(\sigma_A + \sigma_S) ds. \quad (3)$$

Since THz waves are composed of a group of photons, the number density of photons is proportional to the intensity of THz waves, denoted by I . Moreover, we assume that a photon once scattered cannot be scattered twice or more back to the received side. Hence, the variation of the intensity of THz waves, denoted by dI , can be expressed by

$$dI = -I \times \text{Pr}. \quad (4)$$

By integrating the both sides of (4), the intensity of the transmitted THz waves is derived as

$$I = I_0 \exp \left[-(\sigma_A + \sigma_S) \int_{s_0}^{s_1} n(s) ds \right], \quad (5)$$

where I_0 refers to the intensity of the received THz waves, s_0 and s_1 represent the origin and the end of the propagation in the environment, respectively. By considering various types of particles in the atmosphere, such as raindrops,

cloud droplets, atmospheric molecules, and free electrons, the equation (5) can be rewritten as

$$I = I_0 \exp \left[- \sum_{\varsigma} (\sigma_{A|\varsigma} + \sigma_{S|\varsigma}) \int_{s_0}^{s_1} n_{\varsigma}(s) ds \right], \quad (6)$$

where the subscript ς denotes the type of particles. Then, the loss of THz waves caused by collisions with particles, called collisional loss, is defined as

$$\begin{aligned} L_C &\triangleq -10 \log_{10} \frac{I}{I_0} \\ &= 10 \log_{10} e \times \sum_{\varsigma} (\sigma_{A|\varsigma} + \sigma_{S|\varsigma}) \int_{s_0}^{s_1} n_{\varsigma}(s) ds, \end{aligned} \quad (7)$$

where L_C is expressed in the dB scale.

As revealed in (7), the atmospheric attenuation depends on absorption and scattering cross section of each particle and the particle distribution along the propagation path. The former part is related to the properties of medium particles concerned with electrodynamics and quantum mechanics, and the latter one depends on the propagation environment, which can be characterized based on thermodynamics. Therefore, cross section and particle distribution should be respectively studied to achieve a universal attenuation model. Note that the effect of atmospheric turbulence is not considered in the proposed model since the randomly refractive index caused by atmospheric turbulence cannot be deterministically modeled. Instead, the scintillation, beam wandering, and beam spreading effects led by turbulence are investigated statistically in previous studies [22].

B. CROSS SECTION MODEL FOR VARIOUS MEDIUM PARTICLES

The absorption and scattering cross sections for various medium particle depend heavily on the relationship between the particle diameter and the wavelength of the electromagnetic (EM) waves. We can divide all particles in THz space-air-ground channels into four categories in descending order of particle diameter, namely, condensed particles with diameter d approximating wavelength λ , condensed particles with diameter d much less than wavelength λ , single molecules, and free electrons. Their corresponding absorption and scattering models are Mie absorption and scattering, Rayleigh absorption and scattering, molecular absorption and scattering, and Coulomb absorption and Thomson scattering, respectively. Note that we do not consider the case $d \gg \lambda$ since there are no such particles in the atmosphere [23].

1) MIE ABSORPTION AND SCATTERING

For condensed particles with $d \approx \lambda$, the absorption and scattering cross section $\sigma_{A|M}$, $\sigma_{S|M}$, named after Mie absorption and scattering cross section [24], are respectively given by

$$\sigma_{A|M} = \frac{\pi d^2}{2\xi^2} \sum_{m=1}^{\infty} (2m+1) \times \left[\text{Re}(a_m + b_m) - |a_m|^2 - |b_m|^2 \right] \quad (8a)$$

$$\approx \frac{\pi d^2}{2\xi^2} \sum_{m=1}^{m_{\max}} (2m+1) \times \left[\text{Re}(a_m + b_m) - |a_m|^2 - |b_m|^2 \right], \quad (8b)$$

and

$$\sigma_{S|M} = \frac{\pi d^2}{2\xi^2} \sum_{m=1}^{\infty} (2m+1) \left(|a_m|^2 + |b_m|^2 \right) \quad (9a)$$

$$\approx \frac{\pi d^2}{2\xi^2} \sum_{m=1}^{m_{\max}} (2m+1) \left(|a_m|^2 + |b_m|^2 \right), \quad (9b)$$

where $\xi = \pi d/\lambda$ and Mie coefficients a_m , b_m with order m are expressed as

$$a_m = \frac{q^2 j_m(q\xi) [\xi j_m(\xi)]' - \mu_1 j_m(\xi) [q\xi j_m(q\xi)]'}{q^2 j_m(q\xi) [\xi h_m(\xi)]' - \mu_1 h_m(\xi) [q\xi j_m(q\xi)]'}, \quad (10a)$$

$$b_m = \frac{\mu_1 j_m(q\xi) [\xi j_m(\xi)]' - j_m(\xi) [q\xi j_m(q\xi)]'}{\mu_1 j_m(q\xi) [\xi h_m(\xi)]' - h_m(\xi) [q\xi j_m(q\xi)]'}, \quad (10b)$$

where q refers to the complex refractive index of the particle and μ_1 is the ratio of the magnetic permeability of the particle to that of the medium. $j_m(\cdot)$ and $h_m(\cdot)$ denote the spherical functions of Bessel and Hankel of the first kind of order m , respectively. The approximations in (8b) and (9b) are satisfied with an approximated maximum order given by [24]

$$m_{\max} = \left\lceil \xi + 4\xi^{1/3} + 2 \right\rceil, \quad (11)$$

where $\lceil x \rceil$ returns the minimum integer larger than x .

2) RAYLEIGH ABSORPTION AND SCATTERING

When the diameter of condensed particles is much smaller than λ , e.g., $d < \lambda/10$, Rayleigh approximation can be applied into (8b) and (9b), where the Mie coefficients a_m and b_m are all approximately zero except for b_1 . Then, the Rayleigh absorption cross section, denoted by $\sigma_{A|R}$, is derived as [24]

$$\sigma_{A|R} = \frac{\pi d^3}{\lambda} \times \text{Re} \left[i \left(\frac{q^2 - 1}{q^2 + 2} \right) \right], \quad (12)$$

where i is the imaginary unit, and the Rayleigh scattering cross section, denoted by $\sigma_{S|R}$, is [24]

$$\sigma_{S|R} = \frac{2\pi^5 d^6}{3\lambda^4} \times \left| \frac{q^2 - 1}{q^2 + 2} \right|^2. \quad (13)$$

3) MOLECULAR ABSORPTION AND SCATTERING

Due to the small size of single molecules, their absorption effect relies on the radiative transfer theory, as a quantum theory, rather than Mie or Rayleigh theories [18], and the cross sections cannot be acquired through (12) and (13) which are based on classical electrodynamics. In quantum

mechanics, each kind of molecule in gaseous form has intrinsic and discrete vibrational or rotational energy levels. The difference between two energy levels is defined as the energy gap. When the energy gap of a molecule coincides with the energy of an incident photon, the photon can be absorbed and the molecule is transferred from the low energy level to the high energy level. The energy gap of atmospheric molecules generally falls in the THz band and the millimeter wave, such as water vapor for the THz wave and oxygen for the millimeter wave [25]. Therefore, molecular absorption effect due to water vapor is one of the most significant features of THz waves while it is negligible for the lower frequencies [26]. Due to the complicated water vapor energy level, the molecular absorption cross section cannot be derived in a simple form, and only empirical models are available. The Rayleigh scattering cross section for a molecule, denoted by $\sigma_{S|MO}$, is expressed as

$$\sigma_{S|MO} = \frac{8\pi}{3} \left(\frac{2\pi}{\lambda} \right)^4 \alpha^2, \quad (14)$$

where α is the polarizability of the molecule which are on the order of 10^{-30} m^3 [27], [28]. Therefore, the molecular scattering effect can be omitted in THz channels.

4) COULOMB ABSORPTION AND THOMSON SCATTERING

When the particles are free electrons in the ionized environment like plasma, the absorption effect, called Coulomb absorption, is characterized by Coulomb absorption cross section given by [29]

$$\sigma_{A|E} = \frac{\kappa_0 f_{pe}^4}{n_e f^2} \times \frac{1}{\sqrt{1 - \frac{f_{pe}^2}{f^2}}}, \quad (15)$$

where f represents the frequency of the incident EM wave and n_e is the number density of free electrons. The parameter κ_0 and the plasma frequency f_{pe} are respectively given as [29]

$$\kappa_0 = 1.76 \times 10^{-16} \left(\frac{\ln \Lambda_t}{T^{3/2}} \right) [\text{m}^{-1} \text{ Hz}^{-2}], \quad (16)$$

$$f_{pe} = \frac{q_e}{2\pi} \sqrt{\frac{n_e}{\epsilon_0 m_e}}, \quad (17)$$

where T in K refers to temperature of the environment, ϵ_0 represents the permittivity of free space, and q_e and m_e are the charge and the mass of an electron, respectively. Λ_t is the Gaunt factor given as [29]

$$\ln \Lambda_t = \begin{cases} 38.3 + \frac{3}{2} \ln T - \ln f & \text{for } T \lesssim 9.1 \times 10^5 [\text{K}], \\ 45.3 + \ln T - \ln f & \text{for } T \gtrsim 9.1 \times 10^5 [\text{K}]. \end{cases} \quad (18)$$

It is worth noting that, if $f_{pe} > f$, (15) is inapplicable. In such situation the EM waves are unable to transmit through the plasma [29]. For example, the typical electron density in lightning is $10^{24}/\text{m}^3$ [30], and then by substituting it into (17) the plasma frequency is 8.97 THz.

Therefore, THz communications with frequencies lower than 8.97 THz would be blocked in typical lightning. The scattering effect caused by free electrons is called Thomson scattering and characterized by the Thomson scattering cross section defined as [29]

$$\sigma_{S|E} = \frac{8\pi}{3} \left(\frac{q_e^2}{4\pi \epsilon_0 m_e c^2} \right)^2, \quad (19)$$

where c is the speed of light in vacuum.

C. PARTICLE DISTRIBUTION FOR VARIOUS PROPAGATION SCENARIOS

From sea level to outer space, the atmosphere can be roughly divided into three layers based on the altitude, namely, the troposphere with altitude from 0 km to 10 km, the stratosphere with altitude from 10 km to 50 km, the ionosphere with altitude from 50 km to 1000 km, and outer space with altitude larger than 1000 km. As shown in Fig. 1, the particles in different layers have distinct and nonuniform distributions, which can be modeled as follows.

1) GASEOUS MOLECULE DISTRIBUTION

Except water molecules, other atmospheric molecules are in gaseous forms. First, by following the hydro-static equation, the gaseous molecules at an altitude h satisfies [31]

$$dp = -nm_0 g dh, \quad (20)$$

where p is the pressure of the gas, m_0 denotes the mass of one single molecule and g represents the gravitational acceleration. Second, according to the ideal gas law, the pressure, number density, and temperature satisfies [31]

$$p = nk_B T, \quad (21)$$

where k_B refers to the Boltzmann constant and the atmospheric temperature T at altitude h can be modeled as [31]

$$T(h) = \begin{cases} T(0) - \Gamma h, & h \leq 10^4 \text{ m}, \\ T(10^4), & 10^4 \text{ m} < h \leq 5 \times 10^4 \text{ m}, \\ 2000 \text{ K}, & h > 5 \times 10^4 \text{ m}, \end{cases} \quad (22)$$

where the temperature descending coefficient Γ is given by

$$\Gamma = \begin{cases} 0.006 \text{ K/m}, & h < 10^4 \text{ m}, \\ 0, & h > 10^4 \text{ m}. \end{cases} \quad (23)$$

Then, by differentiating the both sides of (21), dp can be expressed as

$$dp = k_B T dn + nk_B dT = k_B T dn - nk_B \Gamma dh. \quad (24)$$

By incorporating (24) into (20), we obtain a differential equation for n as

$$\frac{dn}{n} = -\frac{m_0 g - k_B \Gamma}{k_B T} dh, \quad (25)$$

By integrating both sides of (25), the number density n versus altitude h is given as

$$n(h) = n(0) \exp \left\{ -\frac{(m_0 g - k_B \Gamma)h}{k_B [T(0) - \Gamma h]} \right\}, \quad (26)$$

where $n(0)$ represents the terrestrial number density of the gaseous molecule. The particle distribution expression (26) is valid for all gaseous particles except for water vapor since it could condense and only exists at the altitudes from 0 to 15 km [32].

2) WATER VAPOR DISTRIBUTION

For water vapor, since it may condense into water drops in humid atmosphere, the corresponding distribution should be characterized with respect to two conditions, saturated condition (relative humidity $\geq 90\%$) and unsaturated condition (relative humidity $< 90\%$). By combining the ideal gas law and the Buck equation under the saturated condition [33], the number density of water molecules in gaseous form is given by

$$n_{SW|MO}(h) \approx \frac{550.09}{k_B T(h)} \times e^{\left[19.843 - \frac{T(h)}{234.5}\right] \left[\frac{T(h)-273.15}{T(h)-16.01}\right]}. \quad (27)$$

In the unsaturated condition, the number density of water molecules can be modeled as [32]

$$n_{UW|MO}(h) = n_{UW|MO}(0) \times e^{-h/h_{W0}}, \quad (28)$$

where $h_{W0} = 2000$ m. By utilizing (26), (27), and (28), the spatial distribution of atmospheric molecules can be assessed for various atmospheric conditions. For instance, the environmental parameters including T , n_0 , $n_{UW|MO}(h)$ are different under day and night conditions.

3) CONDENSED PARTICLE AND ELECTRON DISTRIBUTION

For condensed particles and electrons in the atmosphere, including raindrops, cloud droplets and fog droplets, their number densities cannot be modeled according to thermodynamics. Their particle distribution can only be achieved by real-time sensing such as meteorological observation or based on empirical data such as rainfall data. Moreover, the condensed particles only exist in the troposphere as the condensation generally occurs in troposphere but not at higher altitudes [34].

D. TOTAL PROPAGATION LOSS MODEL

Even the propagation environment is a vacuum, due to the inverse-square law and the dependence of the effective area of antenna on frequency, EM wave still suffers from the FSPL given by

$$L_{FP} = 20 \log_{10} \frac{4\pi r}{\lambda}, \quad (29)$$

where r represents the path length and L_{FP} is expressed in the dB scale. By combining (7) and (29), the total propagation loss is expressed as

$$\begin{aligned} L_{tot} &= L_C + L_{FP} \\ &= 10 \log_{10} e \times \sum_{\zeta} (\sigma_{A|\zeta} + \sigma_{S|\zeta}) \int_{s_0}^{s_1} n_{\zeta}(s) ds \\ &\quad + 20 \log_{10} \frac{4\pi r}{\lambda}, \end{aligned} \quad (30)$$

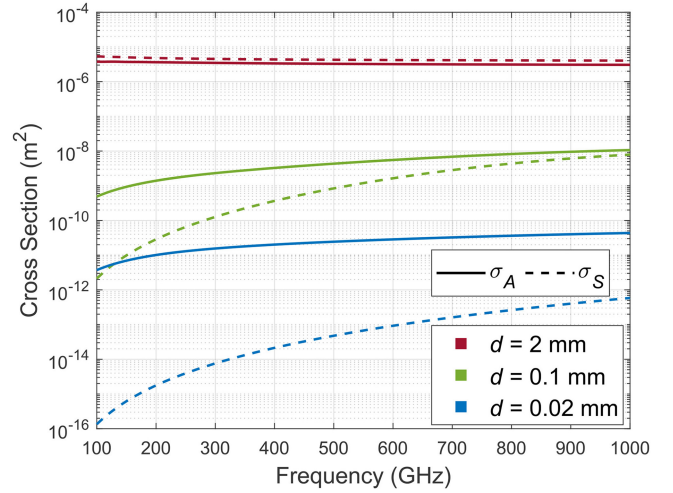


FIGURE 2. The cross sections of water drops with various diameters.

where L_{tot} is expressed in the dB scale. Based on the knowledge about the cross sections of various particles and the particle distribution along the propagation path, the total propagation loss of THz waves can be precisely assessed according to (30).

III. NUMERICAL EVALUATION OF ATTENUATION IN SPACE-AIR-GROUND CHANNEL

In this section, we elaborate the numerical evaluations for the universal attenuation model in space-air-ground channel medium. First, the cross sections for various particles and THz waves with different frequencies are presented. Second, by combining the particle distribution model, the attenuation in various atmospheric layers are elaborated. Finally, our proposed model is compared with ITU models to demonstrate the consistency and improvement based on the state-of-the-art studies.

A. CROSS SECTION

The absorption and scattering cross sections for various particles are numerically evaluated as follows. First, as the most common condensed particle in the atmosphere, the absorption and scattering cross sections of water drops with various diameters are illustrated in Fig. 2, where complex refractive index of water drops is acquired through the double-Debye model. For THz waves from 0.1 to 1 THz, the collision between photons and water drops with $d = 2$ mm, can be modeled by Mie absorption and scattering, where the Mie absorption and scattering cross sections are both on the order of 10^{-6} m². Moreover, we observe that the Mie scattering cross section is slightly larger than the Mie absorption cross section of the water drop with $d = 2$ mm. Moreover, for water drops of $d = 0.02$ mm, the Rayleigh scattering cross section is generally much smaller than the Rayleigh absorption cross section due to $d \ll \lambda$. Therefore, THz waves are attenuated more severely by the scattering effect of water drops rather than their absorption effect.

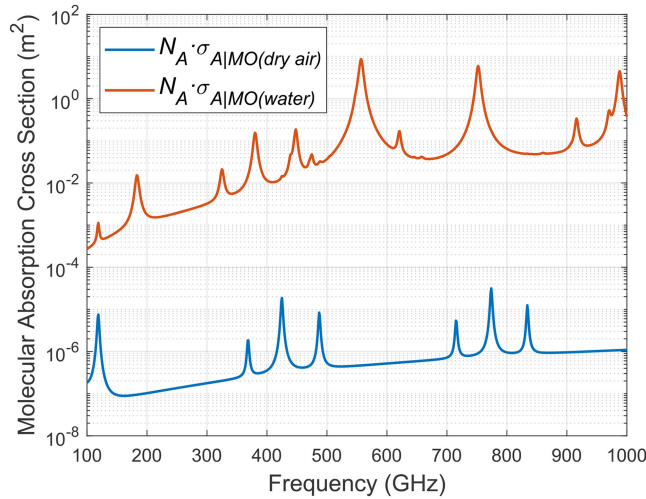


FIGURE 3. The absorption cross section of molecules in gaseous form, where N_A is the Avogadro constant.

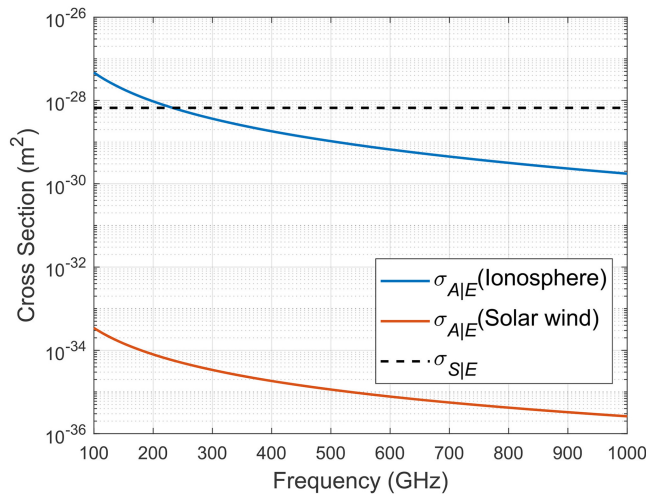


FIGURE 4. The cross sections of free electrons in plasma.

Second, for the molecular absorption and scattering effect, Fig. 3 shows an empirical molecular absorption cross section model for water molecules and dry air molecules, respectively, where the dry air consists of approximately 78.084% nitrogen, 20.948% oxygen, 0.935% argon and 0.033% carbon dioxide [35]. The result suggests that the water vapor dominates the molecular absorption effect in the THz band.

Finally, the cross sections of free electrons are illustrated in Fig. 4. The Coulomb absorption cross section in plasma decreases with the number density of free electrons and frequency, while the Thomson scattering cross section of free electrons shows no dependence with the frequency as expected in (15) and (19). Despite the small cross section of electrons, the number density of electrons can be very high in some extreme scenarios, such as lightning. Therefore, it is necessary to model the influence of electrons in the universal attenuation model.

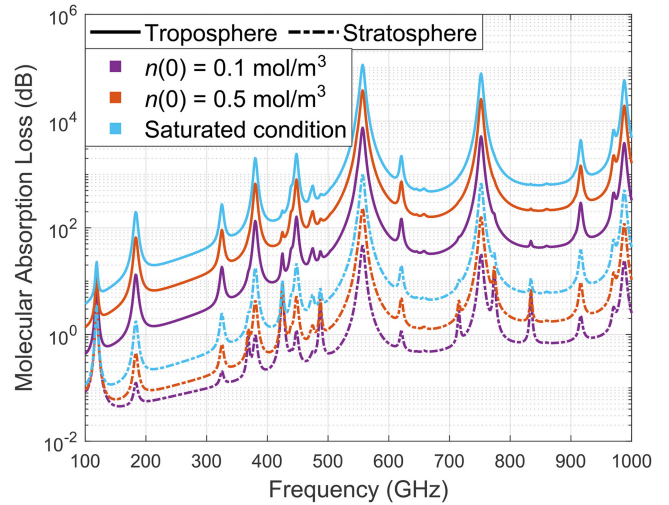


FIGURE 5. The molecular absorption loss in the troposphere and stratosphere, where $T(0) = 298.15$ K.

B. PROPAGATION LOSS IN VARIOUS ATMOSPHERIC LAYERS

After computing the cross sections for various particles, by incorporating the particle distribution model, we can evaluate the attenuation of THz space-air-ground channels.

1) PROPAGATION IN THE TROPOSPHERE

With over half of the mass of the whole atmosphere and the abundant water vapor evaporated from the ocean [36], the troposphere is the most harsh propagation environment for THz waves compared to the other layers of atmosphere. By considering a vertical link across the whole troposphere, i.e., a link of 10 km with elevation angle $\theta = \pi/2$, the molecular absorption loss is shown in Fig. 5. We can observe that in the saturated condition, the THz waves above 320 GHz suffer from severe molecular absorption loss over 100 dB. In the unsaturated condition, such as $n(0) = 0.5$ mol and $n(0) = 0.1$ mol, the molecular absorption loss is much lower than that in the saturated condition due to lower density of water vapor, but the absorption loss is still large at several absorption peaks.

In addition to molecular absorption loss, THz waves may suffer from attenuation from condensed particles in some special weather conditions like rainy, foggy, cloudy days. The absorption and scattering losses caused by condensed particles are illustrated in Fig. 6. In the heavy rain with a rain rate of 98 mm/hr, the diameter and number density of typical raindrops are 2 mm and 10^3 m^{-3} , respectively [37]. For raindrops, the Mie scattering loss is greater than the Mie absorption loss and both of them decrease with an increasing frequency. In heavy fog, the diameter and number density of typical fog droplets are 0.02 mm and 10^8 m^{-3} , respectively [37]. For cloud droplets in cumulonimbus, i.e., a towering vertical cloud associated with heavy rain, the typical diameter and number density are the same as typical fog droplets in heavy fog [37]. Different from the loss of

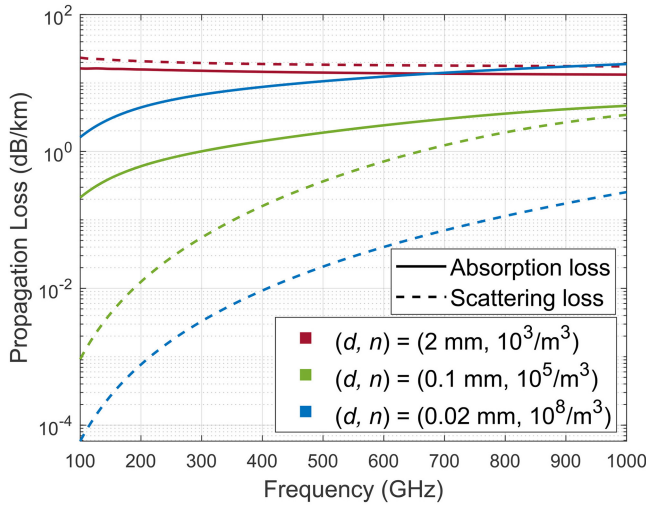


FIGURE 6. The propagation loss caused by condensed particles in the troposphere.

rain, the absorption loss of fog and cloud is much larger than the scattering loss and both of them increase with frequency.

2) PROPAGATION IN THE STRATOSPHERE

Due to the gravity and the decrease of temperature, the water vapor density is lower in the stratosphere and thus smaller molecular absorption loss than in the troposphere as demonstrated in Fig. 5. Moreover, since the aforementioned special weather conditions never happen in the stratosphere, there are no condensed particles or free electrons, and therefore no additional loss in the stratosphere. The remaining attenuation effect is caused by water vapor, and thus the frequency close to the absorption peak should still be avoid for THz links crossing this region.

3) PROPAGATION IN THE IONOSPHERE

In the ionosphere, there are free electrons generated from the ionization of neural molecules by high-energy rays from the Sun and outer space. Fig. 7 illustrates the propagation loss in the ionosphere including the molecular absorption loss, Coulomb absorption loss, and Thomson scattering loss. For the molecular absorption effect, the loss is on the order of 10^{-13} dB/km due to the sparse atmospheric molecules in the ionosphere. For the attenuation led by free electrons, since the highest density of free electrons in the ionosphere is on the order of 10^{11} m^{-3} and the typical temperature of ionosphere is 2000 K [11], the Coulomb absorption loss and Thomson scattering loss for THz waves in the ionosphere are both smaller than 10^{-13} dB/km. Hence, the total propagation loss in the ionosphere is less than 10^{-11} dB/km. Compared with the FSPL over 150 dB, the collisional loss of THz waves in the ionosphere is negligible. Therefore, the FSPL dominates the propagation loss of THz waves in the ionosphere.

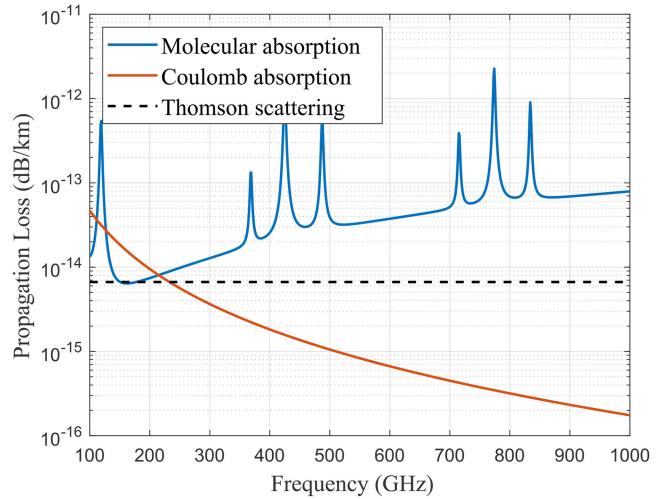


FIGURE 7. The propagation loss in the ionosphere.

4) PROPAGATION IN THE OUTER SPACE

Free electrons in solar wind are the most common particles in the outer space. Their typical temperature and density are 10^5 K and 10^7 m^{-3} , respectively [11]. According to (15) and (19), the propagation loss of THz waves is less than 10^{-24} dB/km. Hence, similar to the conclusion in ionosphere, the FSPL dominates the propagation loss of THz waves in the outer space.

C. COMPARISON WITH EXISTING ITU MODELS

The ITU recommendation models are commonly utilized to predict the propagation loss of the THz waves from 0.1 to 1 THz in the troposphere [15], [16], [17]. However, the ITU models for THz waves are empirical models, which are double-edged swords. On one side, the empirical model is easier to apply compared with theoretical model. On the other side, since empirical models are developed from the data measured in a typical environment, they neglect the difference of various environment and cause inaccuracy between model and reality.

1) COMPARISON WITH ITU-R P.838-3 ON RAIN ATTENUATION

The diameter of raindrops is not a invariant constant and the size distribution of raindrops is called raindrop spectrum. There are various raindrop spectra even at the same rain rate. For instance, the rain above sea consists of raindrops with smaller diameter and higher number density compared with the one above land [23]. Consequently, it is necessary to model the influence of rain spectra to capture its impact. However, ITU-R P.838-3 only incorporates the impact of rain rate and neglects the influence of rain spectra.

As illustrated in Fig. 8, for the same rain rate but with different raindrop diameters 1.5 mm, 2 mm, and 2.5 mm, the propagation losses are different. We see that the rain consisting of raindrops with smaller diameter can induce

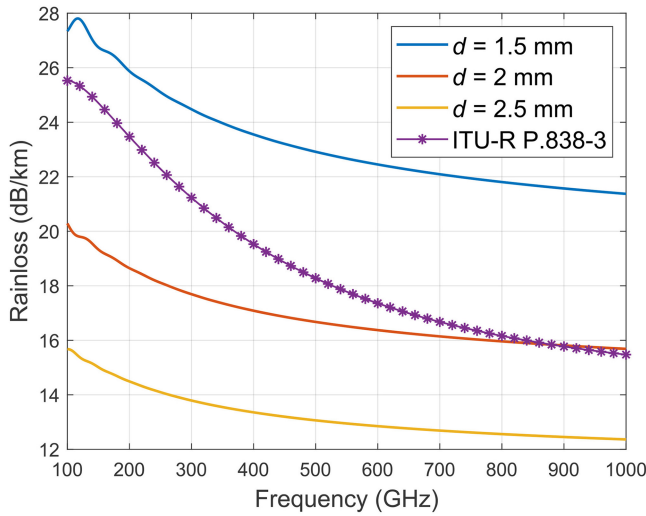


FIGURE 8. The prediction of rain attenuation by the proposed model and ITU-R P.838-3, where the rain is supposed to only consist of raindrops with diameter d and the rain rate is 50 mm/hr.

severer attenuation than that with larger diameter of the same rain rate. Therefore, THz waves suffer from higher propagation loss in the rain above sea than the rain above land, which has not been demonstrated by ITU-R P.838-3. The proposed model can incorporate various rain spectra through varying the parameter n_{ζ} in (30).

In the case without sufficient data on rain spectra, ITU-R P.838-3 can provide a preliminary estimation on the attenuation. While in the case with knowledge on rain spectra, the proposed model can more accurately characterize the propagation loss of THz waves in space-air-ground medium.

2) COMPARISON WITH ITU-R P.676-13 ON MOLECULAR ABSORPTION LOSS

As shown in Fig. 9, molecular absorption loss with elevation angles at 0° , 45° , 90° as well as the model provided in ITU-R P.676-13 are evaluated and compared. It can be observed that for the same propagation distance at 10 km, vertical propagation has a higher molecular absorption loss than horizontal one. This is due to the fact that water vapor density decreases as altitude increases in the atmosphere and thus higher altitude leads to lower absorption loss. Moreover, we see that molecular absorption loss given by the proposed model coincides well with ITU-R P.676-13 at the absorption peaks and slightly differs in the bands between peaks. The difference is caused by the minor difference in the characterization on the number density of these molecules.

Both the proposed model and ITU-R P.676-13 are able to characterize the inhomogeneity of atmospheric molecules and suggest that the molecular absorption loss increases with the propagation distance. However, ITU-R P.676-13 only evaluates the molecular absorption effect while the proposed universal model captures more attenuation effects.

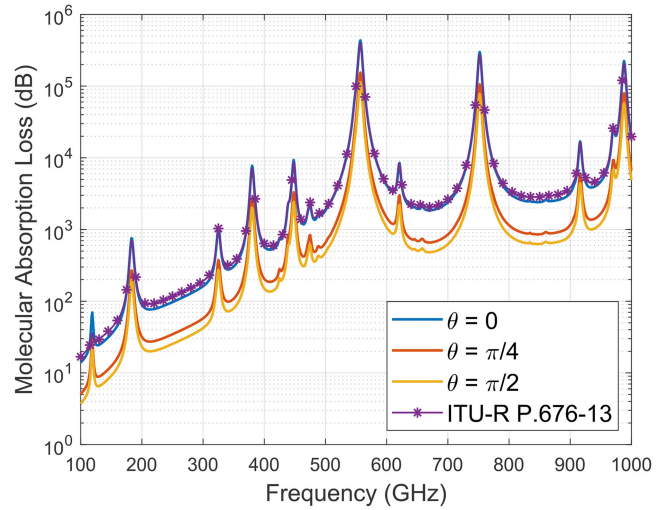


FIGURE 9. The molecular absorption loss with various elevation angles at 0° , 45° , and 90° , based on the proposed model under saturated condition, where the propagation distance is 10 km and $T(0)=298.15$ K.

IV. CONCLUSION

In this paper, we proposed a universal attenuation model for the THz SAGIN, where the absorption and scattering effects induced by condensed particles, molecules, and free electrons are respectively unified. Specifically, the collision model between photons and medium particles is developed, where the cross sections of various particles and the particle distribution are two key parameters characterizing the total propagation loss. On one hand, the cross section is modeled based on electrodynamic property of particle. On the other hand, the particle distribution is analyzed based on thermodynamic features of the propagation environment. Based on the proposed model, the propagation loss of THz waves in the atmosphere and the outer space are numerically assessed. Numerical results indicate that the attenuation effects except free space loss are all negligible at the altitude higher than 50 km while they need to be considered in the atmosphere lower than 50 km. Compared with ITU standards, our proposed model provides a better-explainable and universal attenuation model for THz SAGIN, which could be a theoretic fundamental and a good complementary for current ITU standards on atmospheric attenuation.

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